

Project Initialization and Planning Phase

Date	29 September 2026
Team ID	UNM1659GenieBirds
Project Title	EduGenie - AI-Powered Learning Assistant for Students
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform student learning using artificial intelligence, boosting understanding and study efficiency. It tackles the difficulty of finding clear, personalised help, promising better learning outcomes, less wasted study time, and more confident students. Key features include AI-powered question answering, topic explanations, quizzes with answer checking, summaries and a recommended learning path.

Project Overview	
Objective	The primary objective is to make learning easier by building an AI-powered assistant that gives students instant, clear answers and explanations, tests their understanding and guides what to study next.
Scope	The project covers a web application with a FastAPI backend and a generative AI service, offering Ask a Question, Explain a Topic, Quiz generation with answer checking, Summary and a Recommended Learning Path for students.
Problem Statement	
Description	Students struggle to find simple, personalised explanations when they are stuck. They spend hours searching scattered websites and videos, or depend on costly tuition, and have no quick way to test their understanding.
Impact	Solving this will save study time, improve understanding and exam confidence, and give every student, whatever their budget, access to a personal learning assistant at any hour.
Proposed Solution	
Approach	Using generative AI to answer questions, explain topics at the learner's level, create quizzes and summaries, and suggest a learning path from the student's quiz results, delivered through a simple web interface.
Key Features	- Ask a Question: instant, clear AI-generated answers. - Explain a Topic: simple or detailed explanations with examples.

	- Generate Quiz with Check Answer: practice questions with scoring and feedback. - Summary: short notes for quick revision. - Recommended Learning Path: suggested next topics based on progress. - Secure login and saved study history for each student.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard laptop/PC CPU (Intel i3/i5 or equivalent, 4 cores); no GPU needed as AI runs on a cloud API
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn
Libraries	Additional libraries	pydantic, requests/httpx, python-dotenv, SQLAlchemy, AI provider SDK
Development Environment	IDE	VS Code, Windows PowerShell terminal, Git
Data		
Data	Source, size, format	User accounts, chat/quiz history and scores; SQLite database, under 100 MB, .db (AI-generated content as JSON)